

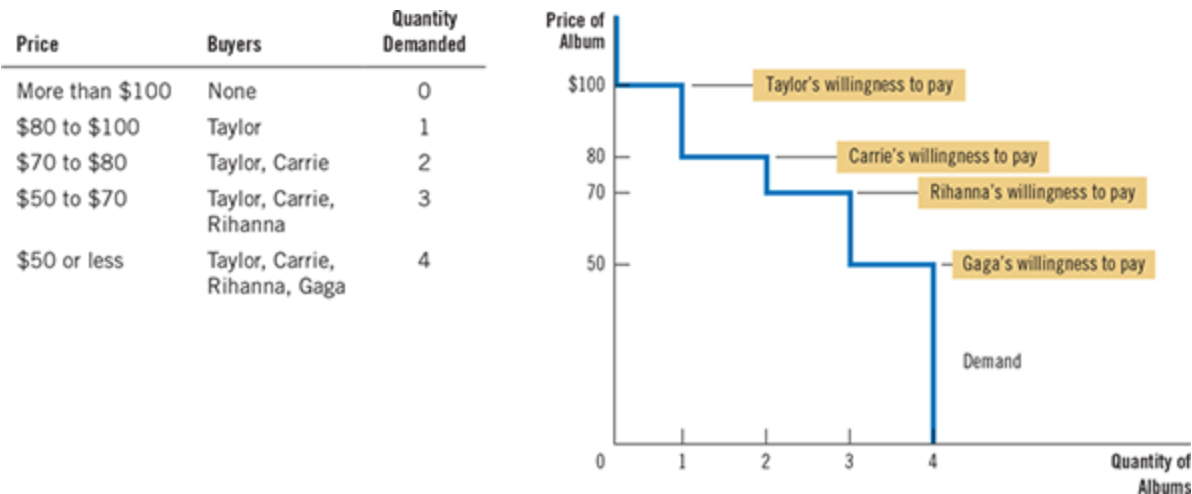
7-1b Using the Demand Curve to Measure Consumer Surplus

Consumer surplus is closely related to the demand curve for a product. To see how they are related, let's continue our example and consider the demand curve for this rare Elvis Presley album.

We begin by using the willingness to pay of the four possible buyers to find the market demand schedule for the album. The table in [Figure 1](#) shows the demand schedule that corresponds to [Table 1](#). If the price is above \$100, the quantity demanded in the market is 0 because no buyer is willing to pay that much. If the price is between \$80 and \$100, the quantity demanded is 1 because only Taylor is willing to pay such a high price. If the price is between \$70 and \$80, the quantity demanded is 2 because both Taylor and Carrie are willing to pay the price. We can continue this analysis for other prices as well. In this way, the demand schedule is derived from the willingness to pay of the four possible buyers.

Figure 1
The Demand Schedule and the Demand Curve

The table shows the demand schedule for the buyers (listed in [Table 1](#)) of the mint-condition copy of Elvis Presley's first album. The graph shows the corresponding demand curve. Note that the height of the demand curve reflects the buyers' willingness to pay.



The graph in [Figure 1](#) shows the demand curve that corresponds to this demand schedule. Note the relationship between the height of the demand curve and the buyers' willingness to pay. At any quantity, the price given by the demand curve shows the willingness to pay of the *marginal buyer*, the buyer who would leave the market first if the price were any higher. At a quantity of 4 albums, for instance, the demand curve has a height of \$50, the price that Gaga (the marginal buyer) is willing to pay for an album. At a quantity of 3 albums, the

demand curve has a height of \$70, the price that Rihanna (who is now the marginal buyer) is willing to pay.

Because the demand curve reflects buyers' willingness to pay, we can also use it to measure consumer surplus. Figure 2 uses the demand curve to compute consumer surplus in our two examples. In panel (a), the price is \$80 (or slightly above) and the quantity demanded is 1. Note that the area above the price and below the demand curve equals \$20. This amount is exactly the consumer surplus we computed earlier when only 1 album is sold.

Figure 2

Measuring Consumer Surplus with the Demand Curve

In panel (a), the price of the good is \$80 and the consumer surplus is \$20. In panel (b), the price of the good is \$70 and the consumer surplus is \$40.

Panel (b) of Figure 2 shows consumer surplus when the price is \$70 (or slightly above). In this case, the area above the price and below the demand curve equals the total area of the two rectangles: Taylor's consumer surplus at this price is \$30 and Carrie's is \$10. This area equals a total of \$40. Once again, this amount is the consumer surplus we computed earlier.

The lesson from this example holds for all demand curves: *The area below the demand curve and above the price measures the consumer surplus in a market.* This is true because the height of the demand curve represents the value buyers place on the good, as measured by their willingness to pay for it. The difference between this willingness to pay and the market price is each buyer's consumer surplus. Thus, the total area below the demand curve and above the price is the sum of the consumer surplus of all buyers in the market for a good or service.

